

Matrix functions and stability

Complete Crouzeix conjecture

Difficulty: extreme **Importance:** broadly interesting**Status:** Solved**Topic:** Matrix-valued polynomial functional calculus

Rating rationale: Historical ratings retained. The uniform bound must survive arbitrary matrix amplification, beyond the scalar functional calculus; it would give sharp bounds for block matrix functions and connect numerical-range estimates to dilation and similarity theory.

Resolution — independently audited 8 October 2026

OpenAI’s [mathematics catalog, family 325](#) reports a proof of the sharp constant-two inequality for finite matrix-valued polynomials, and a stronger version for bounded operators on arbitrary complex Hilbert spaces. Two [independent informal AI-agent audits](#) read all 12 pages of the [direct proof manuscript](#), checked its central positive-kernel and block-matrix arguments, and found no gap. Theorem 1.1 covers all base and coefficient sizes and point or line-segment numerical ranges. Its tensor-factor order differs from the block notation below only by a unitary flip. This establishes the original MF-23 target under the catalog’s informal-audit rule. The stronger [Hilbert-space manuscript](#) was not audited here. The problem ID, canonical path and statement are retained.

At immutable OpenAI revision `adc7f1241b42e322a6451854ab7e4b4c146bf78a`, the project publishes a [Lean formalization scope record](#). Its [finite-matrix proof source](#) declares `OAI.DirectCrouzeix.complete_crouzeix : UniversalBound 2`. The [Comparator challenge](#) defines `UniversalBound` with arbitrary positive matrix sizes, degree and complex coefficients, using the Euclidean operator norm and the supremum over the numerical range. The [formalization catalog](#) maps the declaration to this challenge, and its [Comparator configuration](#) permits only Lean’s standard foundational axioms.

Lean evidence check, 8 October 2026: the pinned proof source declares the exact `UniversalBound 2` target. The [review](#) compared its definitions with the Comparator challenge and scanned the direct proof modules for placeholders and bypasses. The repository publishes a pinned toolchain and dependencies, but no dated successful kernel build or Comparator log and no transitive axiom report were found for the cited revision. This therefore remains **Solved**, not *Lean verified*, under the catalog’s [formal evidence criteria](#). No external human peer review or local Lean rerun is asserted.

Context and notation

All matrices are complex, and all norms are Euclidean operator norms. For $A \in \mathbb{C}^{n \times n}$ define

$$W(A) = \{x^*Ax : x \in \mathbb{C}^n, x^*x = 1\}.$$

If $F(z) = [f_{ij}(z)]_{i,j=1}^m$ has polynomial entries, let $F(A) = [f_{ij}(A)]_{i,j=1}^m$, an $mn \times mn$ block matrix.

Problem statement

Prove or disprove that, for every $n, m \geq 1$, every $A \in \mathbb{C}^{n \times n}$ and every matrix polynomial $F \in \mathbb{C}^{m \times m}[z]$,

$$\|F(A)\|_2 \leq 2 \max_{z \in W(A)} \|F(z)\|_2.$$

This is the complete, rather than scalar, numerical-range spectral-set question. It controls simultaneous polynomial functions and hence block approximation errors with one constant independent of matrix and block sizes.

References

- Michel Crouzeix, [Numerical range and functional calculus in Hilbert space](#), *Journal of Functional Analysis* **244**(2), 668–690 (2007). Original source for the complete conjecture, as explicitly attributed in the next source.
- Per Åhag, Rafał Czyż and Jani Virtanen, [Square Functions, Complete Crouzeix Conjecture in Dimension Three, and the Clouâtre–Ostermann–Ransford conjecture](#), [arXiv:2608.27346v3](#), **9 September 2026**, Introduction, equation (1.2) and **Theorem 1.1**. This current version states the exact target and explicitly distinguishes it from the scalar proofs. It claims the complete bound for $n \leq 3$. Its counterexample reduction leaves only 2×2 matrix polynomials to check when $n = 4$.
- Michel Crouzeix and César Palencia, [The numerical range is a \$\(1 + \sqrt{2}\)\$ -spectral set](#), *SIAM Journal on Matrix Analysis and Applications* **38**(2), 649–655 (2017). The complete universal upper bound is $1 + \sqrt{2}$.

Earlier scope and status check (11 September 2026)

The target is source-stated, not an editorial extension of the scalar conjecture. The August 2026 scalar proof claims by Jin and by Lorist–Schwenninger do not settle it: the September 9 primary manuscript explicitly addresses this distinction. The $n = 3$ result was then a recent preprint claim; the full displayed target remained unresolved even if that claim was accepted. Searches through 11 September 2026 for complete Crouzeix proofs, matrix-valued Crouzeix inequalities and later versions found no full solution. The version was inspected as both HTML and PDF; the inequality is on PDF page 2, immediately before Theorem 1.1.